Disentangling the Black Box: Interpretable Protein-Ligand Docking via Sparse Autoencoders

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Abstract

Protein-ligand docking pipelines based on variational autoencoders (VAEs) achieve strong predictive performance but produce opaque latent representations, while the large number of candidate poses generated per complex imposes substantial computational cost as each undergoes expensive energy minimization and refinement regardless of quality. Physics-informed generative pipelines like BRAID ground predictions in thermodynamic principles, making their learned representations more likely to encode physically meaningful features and making interpretability both more tractable and more scientifically valuable than in end-to-end learned models. We apply sparse autoencoders (SAEs) to the VAE latent space of BRAID, training on 435,069 poses across 36 protein systems. At a 4Å threshold, SAE-derived features filter 16.5% of poor poses prior to refinement while retaining 91.1% of native-like conformations, outperforming BRAID's own energy score. Fraction-of-variance-explained (FVE) analysis shows the top SAE retains 62% of latent variance while still supporting a high-precision filter. Feature analysis reveals sparse features activating exclusively on failed docking attempts corresponding to distinct structural failure modes and providing interpretable windows into what generative models learn about binding physics. As machine learning models take on greater roles in drug discovery, these results demonstrate that SAEs can simultaneously serve as practical computational filters and interpretability tools for building trustworthy, explainable systems.

1 Introduction

Protein-ligand binding governs essentially all biological processes and underlies the majority of small-molecule drug discovery [Dhakal et al., 2022, Sim et al., 2025]. Predicting how a ligand binds to a flexible protein target, the induced fit docking (IFD) problem, remains one of computational biochemistry's most challenging tasks, requiring accurate modeling of coupled conformational changes in both protein and ligand upon binding [Miller et al., 2021]. Most end-to-end machine learning (ML) approaches to this problem, such as DeepDTA, GraphDTA, DGraphDTA, and MGraphDTA, treat binding as a sequence-to-structure mapping and do not explicitly model atomic interactions, sacrificing physical grounding for predictive power [Wu et al., 2025]. In addition, these ML approaches are challenged by limited quality and diversity of training data, a lack of interpretability, and poor generalizability to novel proteins or ligands [Lam and Katritch, 2025, Jiang et al., 2026].

BRAID, Schrödinger and the Friesner Group's upcoming IFD pipeline, combines physics-based simulation with variational autoencoder (VAE) and diffusion generative models to sample conformational space, achieving strong performance in cross-docking benchmarks similar to Mansoor et al. [2024] (Konstantinovsky et al., in preparation). However, this approach generates thousands of candidate poses per complex, each of which undergoes computationally expensive prime energy minimization and algorithmic refinement regardless of quality—a substantial bottleneck when scaling to large compound libraries [Wang et al., 2019].

A second, deeper challenge is interpretability. The VAE compresses protein-ligand poses into low-dimensional latent vectors, but these representations are opaque: it is unclear which directions in latent space encode binding geometry, energetic favorability, or structural quality. This matters not only for scientific understanding but for trust. In drug discovery, black-box predictions carry real consequences, and understanding why a model scores a pose highly is as important as the score itself [Qadri et al., 2025].

Sparse autoencoders (SAEs) have recently emerged as a powerful tool for decomposing opaque neural representations into interpretable, monosemantic features [Templeton et al., 2024, Gao et al., 2024]. Applied to protein language models, SAEs recover biologically meaningful features corresponding to secondary structure, binding sites, and evolutionary conservation [Adams et al., 2025, Simon and Zou, 2025]. Physics-informed pipelines like BRAID are a particularly compelling target for this approach: because the latent space is constrained by thermodynamic priors and explicit atomic interactions, learned features are possibly more likely to correspond to real physical quantities such as energy terms, geometric constraints, and known binding physics [Li et al., 2026, Wu et al., 2024]. This gives us a principled way to validate interpretability claims that is unavailable for end-to-end learned models. While SAEs have been applied to structure-generation models such as RFdiffusion (FoldSAE) and to protein folding representations (ESMFold/ESM2-3B), to our knowledge they have not previously been used to disentangle the latent representations of a physics-informed, molecular-docking generative model for protein–ligand docking.

This work applies TopK SAEs to the VAE latent space of BRAID, addressing both challenges simultaneously. We make three main contributions: (1) We demonstrate that sparse features enable a practical pose-filtering pipeline that flags 16.5% of all test poses for removal while retaining 91.1% of native-like conformations prior to expensive refinement. Because SAE features do not provide predictive lift over raw latents in global classification (auPR 0.719 vs 0.770), we establish that the SAE's value lies in feature monosemanticity and variance retention, supported by FVE analysis showing the top SAE retains 62% of latent variance. (2) We show that raw energy scoring cannot substitute for this filter: a 38% tied mass of failed-minimization sentinel scores prevents energy cutoffs from reaching precise, low-flag operating points entirely, while at higher flag rates where comparison is possible, the SAE achieves far higher precision, recall, and good-pose retention. (3) We isolate sparse features that activate exclusively on structurally failed docking attempts, each corresponding to distinct geometric failure modes—providing a window into what generative models potentially learn about binding physics.

2 Methods

2.1 Dataset

We compiled a dataset of 435,069 molecular docking poses across 36 protein-ligand systems generated by BRAID. Each pose consists of a 30-dimensional VAE latent vector, ligand RMSD to the native structure (excluding hydrogen atoms), and a free energy prediction from physics-based scoring. We use two RMSD-based pose-quality definitions depending on task: native-like poses at RMSD $\leq$ 2Å versus poor poses at RMSD $>$ 2Å (12.3% native-like, following prior work such as Shim et al. [2022]), and good poses at RMSD $\leq$ 4Å versus poor poses at RMSD $>$ 4Å (36.6% native-like) for the conservative filtering task described below. Data was split by case before computing any statistics, yielding approximately 70% train, 15% validation, and 15% test sets. Normalization statistics are computed from the training split only; feature, threshold, and combo selection for the pose-filtering task (Section 2.4) draw on the pooled training and validation splits, with the test split reserved for a single final evaluation.

2.2 TopK Sparse Autoencoder

Following Adams et al. [2025], we train TopK SAEs with $4\times$ expansion: 30D input $\rightarrow$ 120D hidden $\rightarrow$ 30D reconstruction. The TopK operation retains the K largest activations and zeros the rest, enforcing hard sparsity, with reconstruction loss as the training objective. Mean-centering was also applied to the encoder bias initialization, following Simon et al. [2026], who report it as a mitigation for dead TopK features. We trained SAEs across $K \in \{1, 3, 6, 8, 15, 20, 30\}$, 5 seeds per K, for up to 100 epochs using Adam (lr $= 10^{-3}$, batch size 256), selecting the best run per K by validation loss. Rather than fixing a single K, we report results across $K \in \{3, 8, 15, 20, 30\}$ and select the operating K per analysis, since K's effect on both reconstruction fidelity and downstream utility is task- and threshold-dependent (Section 3).

2.3 Pose Quality Classification

To assess whether SAE features provide utility beyond raw latents, we trained logistic regression classifiers on (1) 30D normalized VAE latents (L2 penalty, class-balanced) and (2) 120D sparse activations from each SAE $K \in \{3, 8, 15, 20, 30\}$ (L1 penalty, class-balanced), fit on the training split and evaluated on the held-out test split. We report auPR as the primary metric, since it better handles the class imbalance present at both thresholds than ROC-AUC, which we report secondarily for comparison with prior work.

2.4 Threshold-Based Pose Filtering

We constructed practical pre-refinement filters using a three-way split. For every one of the 120 features per SAE, we fit an F1-maximizing activation cutoff and ranked candidates by precision on a pooled selection set combining the training and validation splits, requiring recall > 0.02 to exclude degenerate near-zero-recall picks. We selected the top 16 candidates by this criterion, then searched all subset unions of those candidates – evaluated on the same pooled selection set – for the union maximizing recall subject to retaining $\geq$80% of native-like poses. The winning union for each SAE is applied exactly once to the held-out test split to report final performance; test-set labels are never used for threshold fitting, candidate ranking, or combo selection. To confirm that filter precision derives from the SAE decomposition rather than raw latent geometry, we apply the identical selection procedure to 500 random unit directions and all 60 signed raw latent axes. Finally, as a physics-based baseline, we apply the same train-select/test-evaluate discipline directly to BRAID's pre-minimization energy score, accounting for its sentinel value assigned to poses where energy minimization fails to converge.

3 Results

3.1 Reconstruction, Classification, and Sparsity

Table 1 reports auPR for bad-pose classification (raw latents vs. SAE K, both thresholds), FVE, and dead-feature rate, all on the test set. Raw latents match or exceed every SAE K on auPR at both thresholds (4Å: 0.770 vs. best 0.750; 2Å: 0.855 vs. best 0.858, effectively tied) – the sparse decomposition provides no consistent predictive lift, so its value lies in filtering (Section 3.2) and interpretability (Section 3.4) rather than classification. FVE grows monotonically with K (33.7%$\rightarrow$90.0%) without a matching auPR gain, and dead-feature rate spikes to 25.8% at K=15 despite mean-centering (Section 2.2, motivated by Simon et al. [2026] as a dead-feature mitigation). This directly explains a downstream oddity: the K=15 filter's winning combo (Section 3.2) includes a feature that is exactly dead on test – zero activation across all 68,614 test poses despite firing 12,679 times in the pooled train+val selection set used to choose it – leaving the deployed 2Å filter effectively 3 live features, not 4.

Table 1: auPR, FVE, and dead-feature rate, raw latents vs. SAE K (test set).

Rep.	auPR (4Å)	auPR (2Å)	FVE	Dead %
Raw latents	0.770	0.855	–	–
SAE K=3	0.709	0.858	33.7%	2.5%
SAE K=8	0.719	0.821	62.0%	1.7%
SAE K=15	0.750	0.846	90.0%	25.8%

3.2 Practical Filtering

Table 2 reports the pooled-train+val-selected combo filter (Section 2.4), applied once to the held-out test split, for $K \in \{3, 8, 15\}$ at both thresholds, alongside the raw-latent-geometry baselines from the identical selection procedure (Section 2.4). K=3 does not survive honest train/validation/test separation: at 2Å its good_kept collapses to 45.8%, and at both thresholds it is statistically indistinguishable from – or, at 2Å, dominated outright by – the raw-axis baseline. K=8 and K=15 are the reproducible choices, adopted as headline operating points at 4Å and 2Å respectively: K=8 flags 16.5% of test poses while retaining 91.1% of native-like conformations (precision 0.803); K=15 flags 15.7% while retaining 93.0% (precision 0.935). Recall is modest at both operating points (0.21, 0.17) – a deliberate precision/good_kept-first design choice appropriate for a pre-refinement filter, where retaining native-like poses matters more than catching every poor one.

At both thresholds, neither raw-latent-geometry baseline family reaches the good_kept $\geq$ 80% target at all: 0 of 65,535 candidate-union combinations clear it for either 500 random unit directions or all 60 signed raw latent axes, so Table 2 reports each baseline's closest achievable point instead. K=8 flags fewer than half as many poses as the best raw-axis baseline while retaining 12 additional points of good_kept (91.1% vs. 79.2%); K=15 shows the same pattern at 2Å (15.7% vs. 27.8% flagged; 93.0% vs. 72.8% good_kept). This independently corroborates adopting K=8/K=15 as headline: sparsity alone does not guarantee an advantage over raw latent geometry, but K=8 and K=15 clearly achieve one.

Table 2: Filtering performance: SAE $K \in \{3, 8, 15\}$ vs. raw-latent-geometry baselines, both thresholds, held-out test set.

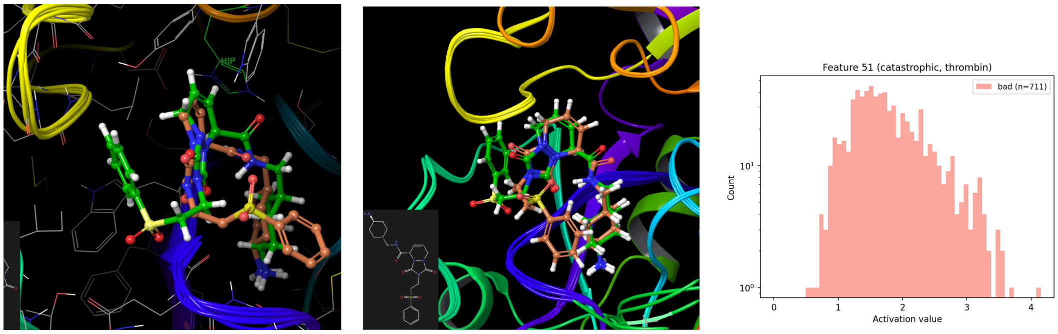
Threshold	Source	% Flagged	Precision	Recall	Good Kept
4Å	SAE K=3	27.3	0.698	0.301	77.5%
4Å	SAE K=8 (headline)	16.5	0.803	0.210	91.1%
4Å	SAE K=15	12.3	0.879	0.171	95.9%
4Å	Random directions (best)	33.7	0.605	0.321	63.6%
4Å	Raw latent axes (best)	27.8	0.727	0.319	79.2%
2Å	SAE K=3	38.1	0.791	0.353	45.8%
2Å	SAE K=8	19.7	0.884	0.204	84.4%
2Å	SAE K=15 (headline)	15.7	0.935	0.172	93.0%
2Å	Random directions (best)	33.7	0.824	0.325	59.7%
2Å	Raw latent axes (best)	27.8	0.857	0.280	72.8%

3.3 Comparison to BRAID's Energy Score

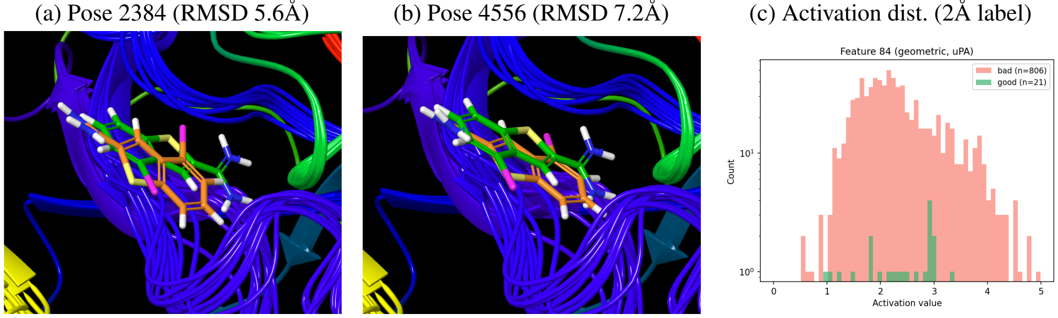
As a physics-based baseline, we apply the identical train-select/test-evaluate discipline directly to BRAID's pre-minimization energy score. 38.4% of test poses at both thresholds are tied at BRAID's sentinel value (energy = 10000), assigned when minimization fails to converge; because this is a tied point mass, no energy threshold can flag fewer than 38.4% of poses regardless of tuning. Both SAE headline operating points (16.5% at 4Å, 15.7% at 2Å) fall below this floor and are therefore structurally unreachable by energy thresholding – not merely outperformed, but inaccessible by construction. At its most conservative achievable point (the sentinel-only filter, 38.4% flagged), energy reaches 78.9% good_kept at 4Å and 78.2% at 2Å, both well below the SAE filters' 91.1%/93.0% despite flagging more than double the poses. An unconstrained best-F1 energy threshold performs even worse, degenerating to flagging 100% of poses – energy alone does not separate pose quality well enough to beat a trivial "flag everything" baseline.

3.4 Interpretable Features

Table 2's K=8 combo filter includes two features with sharply contrasting structural signatures. Feature 51 fires on 711 test poses, 99.3% of which come from a single thrombin case pair [Di Cera et al., 1995] (`throm_1ae8_throm_1d9i`, 706/711; a second thrombin pair accounts for the remaining 5), at precision 0.809 with a sentinel (crashed-minimization) rate of 86.4% against a 38.4% population



(a) Pose 2384 (RMSD 5.6Å) (b) Pose 4556 (RMSD 7.2Å) (c) Activation dist. (2Å label)



(d) Pose 11084 (RMSD 5.7Å) (e) Pose 11377 (RMSD 5.0Å) (f) Activation dist. (2Å label)

Figure 1: Feature 51 (top, thrombin, `throm_1ae8_throm_1d9i`) and Feature 84 (bottom, uPA, `upa_2vio_upa_1c5w`), both members of the deployed K=8 4Å combo filter. Left/middle: highest-activating test poses, docked pose (green) overlaid against a reference conformation (orange). Feature 51's poses show the ligand's bulky substituent rotated toward Ala190 instead of the native-contacting Leu99, near the catalytic HIP (protonated histidine) residue; the multiple colored backbone traces reflect BRAID's induced-fit receptor-loop sampling (confirmed present in both good and bad poses for this case, not a bad-pose-specific artifact – see Discussion). Feature 84's poses show a consistently-oriented ligand translationally shifted within the binding pocket. Right: activation distribution (active poses only, log-scale count) split by the 2Å native-pose criterion – Feature 51 activates exclusively on non-native poses (0/711 good); Feature 84 activates on non-native poses at 806/827 (97.5%), with false positives visibly sparse relative to the bad-pose mass.

baseline. Under the stricter, standard 2Å native-pose criterion, precision is perfect: all 711 flagged poses are non-native (0 false positives). Its highest-activating poses (Figure 1) show a consistent rotational failure: the ligand's sterically bulky substituent, which should point toward Leu99, instead points toward Ala190 – both residues lining the S1/S2 subsites adjacent to the catalytic triad [Bode et al., 1989]. This misorientation is severe enough that minimization crashes outright for most flagged poses, consistent with the elevated sentinel rate.

Feature 84 tells a different story. It fires on 827 poses, 94.3% concentrated in a single urokinase (uPA) case pair [Spraggon et al., 1995] (`upa_2vio_upa_1c5w`, 780/827) – the same catalytic serine-protease fold family as thrombin – at precision 0.911 with a sentinel rate of 28.5%, *below* the population baseline; at the 2Å native-pose criterion, precision rises to 0.975 (806/827). Its highest-activating poses show the ligand correctly oriented but translationally displaced within the binding pocket; energies converge to real, physically stable values (clustered around $-9{,}300$ to $-9{,}700$) rather than the sentinel value, consistent with a milder geometric error that minimization can still resolve. On the 20 most confidently native poses for this case (RMSD $<$0.4Å), feature 84's activation is exactly 0.0000 – the feature is silent on good poses rather than weakly active everywhere.

Both features are live members of the deployed K=8 combo filter (Section 3.2). Together they illustrate a rotational-vs-translational split in the geometric errors the SAE decomposes into distinct, monosemantic features: severe rotational misorientation that crashes minimization (Feature 51), and milder translational displacement that minimization tolerates (Feature 84) – consistent with the monosemanticity hypothesis that sparse features encode qualitatively distinct failure phenomena rather than a single distributed correlate of pose quality [Templeton et al., 2024, Adams et al., 2025, Clark et al., 2026, Bricken et al., 2023].

4 Discussion

4.1 Sparse Features as Targeted Structural Detectors

The central finding of this work is not that SAEs improve classification accuracy – raw latents match or exceed every SAE K on auPR at both thresholds (Section 3.1). The more interesting result is what the sparse features are: precise, case-specific structural-failure detectors, and a practical filter that raw latent geometry and BRAID's own energy score cannot match.

The K=8 filter's constituent features split the bad-pose manifold into qualitatively distinct failure types. Feature 51 fires overwhelmingly on poses where minimization crashes outright (86.4% sentinel rate vs. 38.4% baseline); its highest-activating poses show a bulky ligand substituent misrouted toward Ala190 instead of the native-contacting Leu99, adjacent to the catalytic triad. Feature 84 shows the opposite signature – a below-baseline sentinel rate (28.5%) and real, converged energies – and flags poses whose ligand is correctly oriented but translationally displaced. Severe misorientation that crashes minimization versus milder displacement that minimization tolerates is consistent with the monosemanticity hypothesis that sparse features encode qualitatively distinct phenomena rather than a single distributed correlate of pose quality [Templeton et al., 2024, Adams et al., 2025, Clark et al., 2026, Bricken et al., 2023].

We searched for the mirror-image case – features activating preferentially on native-like poses – using the identical train-select/test-evaluate discipline across both headline models. None approached the precision or concentration of the failure-mode detectors above: the best 4Å candidate reached 55.3% precision at 1.6% recall, and the best 2Å candidate reached 22.2% precision at 11.1% recall, both barely above their respective population baselines (36.6%, 14.7%). This asymmetry suggests the SAE more readily isolates specific, localized ways a pose fails than the more diffuse geometric requirements of a correct one.

Two checks rule out simpler explanations. The random-direction and raw-axis baselines (Section 3.2) show 0 of 65,535 candidate-union combinations reach the good_kept $\geq$ 80% target that K=8/K=15 clear easily, so the SAE decomposition – not linear latent geometry – drives the precision, and K=3's failure to separate from this baseline underscores that sparsity alone is not sufficient. Separately, we tested whether the induced-fit receptor-loop flexibility visible in Feature 51's flagged poses (Figure 1) is itself diagnostic; it is not – the same conformational spread is present in the case's most confidently native poses, ruling out an otherwise tempting claim that the feature tracks receptor flexibility rather than ligand geometry.

That Feature 51 and Feature 84 each concentrate in a single case pair (99.3% and 94.3%) is evidence the SAE learns protein-specific failure signatures a global classifier would smear across cases – consistent with raw latents matching SAE auPR overall despite this case-level specificity. SAE-based filters are therefore most valuable when the target protein family is known, complementary to case-agnostic scoring rather than a replacement for it. Recall is modest at both headline operating points (0.21, 0.17), a deliberate precision-first design choice rather than a ceiling on detectability; generalization beyond the 36 systems studied here, and beyond the two failure modes highlighted, requires further validation, and activation steering is a natural next step for closing the loop between interpretability and generative model improvement [Zarzecki et al., 2025].

5 Conclusion

We have shown that sparse autoencoders can decompose the latent space of a physics-informed docking pipeline into interpretable, structurally grounded, case-specific features – high-precision detectors for distinct failure modes that fire rarely and in ways mappable to physical phenomena. At the 4Å threshold, the K=8 combo filter removes 16.5% of poor poses while retaining 91.1% of native-like conformations, beating both a raw-latent-geometry baseline and BRAID's own energy score, which is structurally incapable of reaching this operating point at all; the K=15 filter achieves the analogous result at 2Å (15.7% flagged, 93.0% retained). SAEs offer a path toward docking pipelines that are not only predictive but mechanistically understood – systems whose failures can be diagnosed, explained, and, eventually, corrected.

Broader Impact This work aims to reduce computational cost and improve interpretability in structure-based drug discovery pipelines by identifying and filtering low-quality docking poses before expensive refinement, and by surfacing which physical failure modes a generative model has and has not learned to represent. Because the method filters and interprets existing poses rather than generating novel molecules, it does not meaningfully lower the barrier to designing novel toxic compounds, the primary dual-use concern in ML-for-drug-discovery work. A more realistic risk is uneven performance: our filters are highly case-specific (Section 3.4), so their precision guarantees may not transfer to protein families outside the 36 systems studied here, and over-reliance on such filters without periodic validation against well-characterized targets could disproportionately deprioritize research on rarer or less-studied disease targets.

References

Etowah Adams, Liam Bai, Minji Lee, Yiyang Yu, and Mohammed AlQuraishi. From mechanistic interpretability to mechanistic biology: Training, evaluating, and interpreting sparse autoencoders on protein language models. *bioRxiv*, 2025.

W. Bode, I. Mayr, U. Baumann, R. Huber, S. R. Stone, and J. Hofsteenge. The refined 1.9 å crystal structure of human alpha-thrombin: interaction with D-Phe-Pro-Arg chloromethylketone and significance of the Tyr-Pro-Pro-Trp insertion segment. *The EMBO Journal*, 8(11):3467–3475, 1989.

T. Bricken et al. Towards monosemanticity: Decomposing language models with sparse autoencoders. *Transformer Circuits Thread*, 2023. URL `https://transformer-circuits.pub/2023/monosemantic-features/index.html`.

J. D. Clark, T. J. Dean, and D. Shukla. Learning physical interactions to compose biological large language models (LLMs). *Communications Chemistry*, 9:81, 2026.

A. Dhakal, C. McKay, J. J. Tanner, and J. Cheng. Artificial intelligence in the prediction of protein–ligand interactions: Recent advances and future directions. *Briefings in Bioinformatics*, 23 (1):bbab476, 2022.

Enrico Di Cera, Enriqueta R. Guinto, Alessandro Vindigni, Quoc D. Dang, Youhna M. Ayala, Meng Wuyi, and Alexander Tulinsky. The Na+ binding site of thrombin. *Journal of Biological Chemistry*, 270(38):22089–22092, 1995.

L. Gao, T. D. la Tour, H. Tillman, G. Goh, et al. Scaling and evaluating sparse autoencoders. *arXiv preprint arXiv:2406.04093*, 2024.

J. Jiang, D. Li, G. Wang, and G.-W. Wei. Recent advances in machine learning predictions of protein-ligand binding affinities. *Current Opinion in Structural Biology*, 96:103193, 2026.

J. H. Lam and V. Katritch. Navigating structure-based drug discovery with emerging innovations in physics- and knowledge-based approaches. *Npj Drug Discovery*, 2(1):29, 2025.

Y. Li, R. Zhan, J. Rao, M. Liu, et al. Structure-informed machine learning for drug discovery: A task-centric perspective. *Briefings in Bioinformatics*, 27(1):bbag081, 2026.

S. Mansoor, M. Baek, H. Park, G. R. Lee, and D. Baker. Protein ensemble generation through variational autoencoder latent space sampling. *Journal of Chemical Theory and Computation*, 20 (7):2689–2695, 2024.

E. B. Miller, R. B. Murphy, D. Sindhikara, K. W. Borrelli, et al. Reliable and accurate solution to the induced fit docking problem for protein–ligand binding. *Journal of Chemical Theory and Computation*, 17(4):2630–2639, 2021.

Y. A. Qadri, S. Shaikh, K. Ahmad, I. Choi, et al. Explainable artificial intelligence: A perspective on drug discovery. *Pharmaceutics*, 17(9):1119, 2025.

H. Shim, H. Kim, J. E. Allen, and H. Wulff. Pose classification using three-dimensional atomic structure-based neural networks applied to ion channel–ligand docking. *Journal of Chemical Information and Modeling*, 62(10):2301–2315, 2022.

J. Sim, D. Kim, B. Kim, J. Choi, and J. Lee. Recent advances in AI-driven protein-ligand interaction predictions. *Current Opinion in Structural Biology*, 92:103020, 2025.

E. Simon and J. Zou. InterPLM: discovering interpretable features in protein language models via sparse autoencoders. *Nature Methods*, 22:2107–2117, 2025.

Elana Simon, Etowah Adams, and James Zou. On the relationship between activation outliers and feature death in sparse autoencoders. *arXiv preprint arXiv:2605.31518*, 2026.

Glen Spraggon, Christopher Phillips, Ursula K. Nowak, Christopher P. Ponting, Derek Saunders, Christopher M. Dobson, David I. Stuart, and E. Yvonne Jones. The crystal structure of the catalytic domain of human urokinase-type plasminogen activator. *Structure*, 3(7):681–691, 1995.

A. Templeton, T. Conerly, J. Marcus, J. Lindsey, et al. Scaling monosemanticity: Extracting interpretable features from Claude 3 Sonnet. *Transformer Circuits Thread*, 2024. URL `https://transformer-circuits.pub/2024/scaling-monosemanticity/index.html`.

E. Wang, H. Sun, J. Wang, Z. Wang, et al. End-point binding free energy calculation with MM/PBSA and MM/GBSA: Strategies and applications in drug design. *Chemical Reviews*, 119(16):9478–9508, 2019.

M.-H. Wu, Z. Xie, and D. Zhi. A folding-docking-affinity framework for protein-ligand binding affinity prediction. *Communications Chemistry*, 8(1):108, 2025.

Y. Wu, B. Sicard, and S. A. Gadsden. Physics-informed machine learning: A comprehensive review on applications in anomaly detection and condition monitoring. *Expert Systems with Applications*, 255:124678, 2024.

W. Zarzecki, P. Szymczak, E. Szczurek, and K. Deja. FoldSAE: Learning to steer protein folding through sparse representations. *arXiv preprint arXiv:2511.22519*, 2025.

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Anonymization pending – see note above. Training/analysis code will be released; an anonymized version should be linked here for the double-blind submission.

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